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Synergizing causation and effectuation for innovation in Chinese new ventures

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ABSTRACT

Innovation often involves high uncertainty, making it difficult for a single entrepreneurial decision-making logic to navigate effectively. Grounded in resource orchestration theory, we advance the discussion on entrepreneurial decision-making logics by examining the interplay between causation and effectuation, and their interactive impacts on innovation within new ventures. Using a multi-wave survey of 404 founding team members from 91 new ventures, we find that new venture innovation is enhanced by the synergy of causation and effectuation. Furthermore, the boundary-spanning behavior of founding teams mediates the relationship between the synergy and new venture innovation, and founding team trust positively moderates this mediated relationship. These insights contribute to understanding the entrepreneurial decision-making logics and strategic behaviors of founding teams as they pursue innovation.

KEYWORDS

Synergy of causation and effectuation; resource orchestration; team trust; team boundary-spanning behavior; new venture innovation

Introduction

China's entrepreneurship ecosystem, nurtured by robust government support, a thriving digital economy, and strategic emphasis on technological advancement, has emerged as one of the world's most vibrant hubs for new venture creation. Within this dynamic landscape, new venture innovation—the firm's willingness to embrace change and implement novel ideas, processes, or products in pursuit of strategic renewal (Calantone et al., 2002)—plays a pivotal role in driving competitive advantage and long-term growth for new ventures, especially in turbulent environments (Adomako et al., 2024; Kim & Mauborgne, 2014). Despite its significance, realizing innovation poses severe challenges for new ventures. The inherently uncertain and resource-intensive nature demands capabilities that these firms often lack, such as adequate financial capital, experienced leadership, structured innovation routines, and access to external networks (Deligianni et al., 2022; Drnevich & West, 2023; Li et al., 2025). While the

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entrepreneurial infrastructure—such as incubators, accelerators, and other support institutions—is often seen as a remedy for these constraints (Aerts et al., 2007; Mian et al., 2016), their effectiveness is often hampered by issues such as heterogeneous tenant composition, weak selection mechanisms, and insufficient monitoring (Dvouletý et al., 2018). Consequently, the burden of strategic resource orchestration often falls on founding teams, who must proactively manage internal and external constraints to foster innovation (Sirmon et al., 2007).

Entrepreneurial decision-making logics—particularly causation and effectuation—provide a critical theoretical lens for understanding how founding teams orchestrate resources, navigate uncertainty, and pursue entrepreneurial opportunities (Cowden et al., 2024; Maine et al., 2015; Sarasvathy, 2001). *Causation* is characterized by structured planning, predictive analysis, and resource optimization within stable contexts (Chandler et al., 2011), and *effectuation* emphasizes flexibility, experimentation, and leveraging contingencies in uncertain settings (Deligianni et al., 2022). Rather than being mutually exclusive (Brettel et al., 2012; Dew et al., 2009), causation and effectuation are increasingly seen as complementary. As Sarasvathy (2001, p. 245) argued, “both causation and effectuation are integral parts of human reasoning that can occur simultaneously, overlapping and intertwining over different contexts of decisions and actions.” This recognition has led to the concept of the synergy between causation and effectuation. The synergy refers to the integrated use of causation and effectuation, where combined application yields greater benefits than using either one alone (Arya, 2002; Feldman & Hernandez, 2022; Smolka et al., 2018; Tanriverdi & Venkatraman, 2005; Yu et al., 2018). Unlike a static blend, the emphasis of integration varies depending on tasks, contexts, or organizational levels (Galkina & Lundgren-Henriksson, 2017; Reymen et al., 2017; Yang & Gabrielsson, 2017).

This synergy is increasingly evident not only in theoretical discussions but also in practical applications. A vivid example can be found in Xiaomi’s founding journey (2010–present). Initially, the founding team employed causation, conducting comprehensive market analyses to identify an underserved segment for premium-yet-affordable smartphones. They then designed and optimized supply chain and distribution strategy. However, when faced with manufacturing constraints and rapidly shifting consumer preferences, the team strategically pivoted to effectual processes. They leveraged existing engineering capabilities, co-created product features through online community engagement, and progressively refined their flagship Mi Mix series through rapid iteration cycles. This deliberate synthesis of predictive planning and adaptive execution enabled Xiaomi to circumvent traditional industry barriers and ultimately achieve both technological innovation and global market penetration.

However, integrating causation and effectuation poses inherent challenges due to their contrasting principles (Galkina et al., 2022). The conceptual differences between causation and effectuation (Braun & Sieger, 2021; de Villiers Scheepers et al., 2025; Dew et al., 2009) create cognitive and operational obstacles for founding teams attempting to reconcile strategic direction with adaptive action (Lanivich et al., 2023). Cognitively, entrepreneurs operate under bounded rationality (Simon, 1972), making it difficult to manage divergent entrepreneurial decision-making logics simultaneously. Personal experience and stable cognitive styles often lead to a preference for one approach, hindering integration (Festinger, 1957; Nummela et al., 2014). Switching between causation and effectuation may cause discomfort or overwhelm (Yang et al., 2019), undermining decision quality. Operational tensions arise from inconsistent information, conflicting team orientations, and misaligned stakeholder expectations (Galkina et al., 2022). These frictions can dilute synergy benefits, cause coordination breakdowns, and hinder timely responses in uncertain markets. Prior research even cautions that excessive integration can produce diminishing returns, as dual implementation without coherence may lead to inefficiencies or strategic ambiguity (Andriopoulos & Lewis, 2009; Hansen et al., 2019).

While existing literature acknowledges these concerns, studies investigating the interplay between causation and effectuation and the conditions under which their synergy may succeed are scarce. Most studies focus on the antecedents of innovation—such as resources (Clausen & Molden, 2024; Eisenhardt & Schoonhoven, 1996), networks (Ricciardi et al., 2025; Stam & Elfring, 2008; Wang et al., 2024), team characteristics (Fox et al., 2023; Huang et al., 2023), and entrepreneurial orientation (Kraus et al., 2023; Lumpkin & Dess, 1996)—but overlook the interplay of entrepreneurial decision-making logics (Galkina et al., 2022). Qualitative research has begun to map how entrepreneurs integrate causation and effectuation in response to critical events (Galkina & Lundgren-Henriksson, 2017; Reymen et al., 2015), but robust empirical studies remain limited (Alzamora-Ruiz et al., 2021). This research gap is theoretically significant because it impedes our understanding of how decision-making logic duality can be strategically orchestrated to foster new venture innovation in uncertain environments. In practical terms, it holds immense practical importance: in real-world high-velocity contexts, entrepreneurs must not only use both causation and effectuation but also resolve their tensions to achieve synergy.

Our article investigates this critical research question: *How do founding teams effectively integrate causation and effectuation to drive innovation in new ventures?* Grounded in resource orchestration theory, which underscores the structuring, bundling, and leveraging of resources for competitive advantage (Helfat & Peteraf, 2003; Sirmon et al., 2011), we propose boundary-spanning behavior as a pivotal mechanism. At the team level, this refers to the actions of

founding team members to coordinate, communicate, and exchange resources with parties in the external environment (Marrone, 2010). In China's resource-constrained yet dynamic market, boundary-spanning behavior enables teams to integrate structured planning with adaptive action, enhancing their capacity to translate decision logic synergy into innovation outcomes. Moreover, we identify founding team trust as a critical moderator. Defined as shared perception of each other's reliability and capability (De Jong & Elfring, 2010), team trust mitigates internal tensions arising from divergent decision preferences and facilitates smooth coordination—essential for the founding team's effective integration of causation and effectuation. Using data from 404 founding team members across 91 new ventures, we empirically test and validate the theoretical model proposing that the synergy of causation and effectuation enhances new venture innovation through the founding team's boundary-spanning behavior, and that this relationship is moderated by founding team trust.

This study makes several important contributions. First, it enriches entrepreneurial decision-making research by theorizing how the synergy of causation and effectuation fosters new venture innovation, moving beyond broad performance outcomes to this critical focus. Specially, the study explores the conditions under which the synergy between causation and effectuation promotes innovation and uncovers boundary-spanning behavior as a vital mechanism. It demonstrates how founding teams blend structured planning with adaptive flexibility to orchestrate resources effectively, addressing a gap in understanding how causation and effectuation interplay. Second, the study extends resource orchestration theory into entrepreneurial contexts, demonstrating that integrating external resources through boundary-spanning refines orchestration beyond internal processes. Third, the study repositions founding team trust as a strategic role from mere team cooperation to a key driver of innovation. This perspective emphasizes the importance of trust in fostering synergy of causation and effectuation in driving innovation. Overall, our study offers a framework that enhances scholarly understanding of the interplay between causation and effectuation. It also provides valuable guidance to practitioners in addressing the tension between causation and effectuation while maximizing their synergy in innovation.

Theoretical background and hypothesis development

The synergy of causation and effectuation and new venture innovation

Drawing on resource orchestration theory (ROT), we argue that the synergy of causation and effectuation constitutes a strategic resource orchestration that significantly enhances innovation in new ventures. ROT posits that strategic managerial actions, such as structuring,

bundling, and leveraging of resources—are critical determinants of a firm's competitive advantage and innovation performance (Chirico et al., 2024; Sirmon et al., 2007). Structuring involves the effective identification and accumulation of resources, bundling refers to the strategic integration of resources to create new capabilities, and leveraging refers to the strategic application of these integrated resources to exploit opportunities (Sirmon et al., 2011).

Causation, rooted in predictive reasoning and deliberate planning, equips teams with a structured approach to resource management. It emphasizes goal setting, systematic resource allocation, and optimization, fostering efficiency and reducing uncertainty through well-defined strategic pathways (Chandler et al., 2011). For instance, causation enables teams to identify critical resources early, align them with predefined objectives, and streamline operations—a process that mirrors ROT's structuring and bundling phases. In contrast, effectuation champions adaptability and resourcefulness, empowering teams to creatively recombine limited resources and respond swiftly to unforeseen opportunities or challenges (Deligianni et al., 2022; Fisher, 2012). It aligns with ROT's leveraging phase, as it encourages the dynamic application of resources in unpredictable environments.

While causation and effectuation each offer distinct advantages for innovation, their integration yields a synergistic effect that transcends their individual contributions (Smolka et al., 2018). Causation establishes a stable foundation of planning and control, mitigating the risks of over-flexibility inherent in effectuation. Conversely, effectuation introduces agility and creativity, countering the rigidity that causation alone might impose (Brettel et al., 2012; Brinckmann et al., 2010). This synergy enables founding teams to orchestrate resources with both precision and adaptability, effectively bridging the gap between systematic efficiency and opportunistic innovation (Furlotti et al., 2020). For example, a team employing causation might secure key technologies through structured investments. Effectuation, on the other hand, could guide the team to pivot these assets toward unexpected market niches, thereby enhancing innovation through complementary strengths.

This synergy is especially valuable in resource-constrained and uncertain contexts, which are common in new ventures. By integrating foresight from causation with responsiveness from effectuation, teams can dynamically adjust their innovation strategies to address emerging threats and seize fleeting opportunities (Maine et al., 2015; Sarasvathy et al., 2014). This orchestration not only mitigates the limitations of each entrepreneurial decision-making logic, such as causation's potential inflexibility or effectuation's lack of direction, but also amplifies their combined capacity to generate novel solutions. Consequently, we posit that the integration of causation and effectuation may represent a strategic capability that helps new ventures optimize resource orchestration and drive their innovation.

Building on this theoretical foundation, we propose the following hypothesis:

Hypothesis 1: The synergy of causation and effectuation employed by the founding team positively influences new venture innovation.

The mediating effect of boundary-spanning behavior of the founding team

Causation–effectuation synergy enables founding teams to combine systematic planning with adaptive experimentation (Yu et al., 2018). This synergy has the potential to unlock the full innovation potential of new ventures (Alzamora-Ruiz et al., 2021). However, achieving this full potential requires more than simply internal resource orchestration—it may also involve active engagement in boundary-spanning behaviors (Chirico et al., 2024; Katila & Ahuja, 2002). Defined as actions to establish linkages and manage interactions with parties in the external environment (Marrone, 2010), boundary-spanning behaviors play a crucial role in bridging the internal innovation processes with external resources and opportunities that are critical for new venture innovation (Leicht-Deobald et al., 2025). By leveraging causation and effectuation in tandem, founding teams may be more likely to engage in boundary-spanning behaviors, thereby enhancing their chances of driving new venture innovation.

Causation, emphasizing structured planning and predictive control, drives teams to systematically identify resource gaps and pursue targeted external partnerships (Chandler et al., 2011). In contrast, effectuation's adaptability enables teams to capitalize on serendipitous discoveries and creatively leverage external networks to overcome resource constraints and uncertainty (Galkina & Atkova, 2020; Galkina & Lundgren-Henriksson, 2017). These dynamics collectively enhance the team's ability to accumulate external knowledge, cultivate partnerships, and mobilize diverse resources, which are essential inputs for innovation (Gibson & Dibble, 2013; Troilo et al., 2014). Boundary-spanning thus potentially functions as a crucial conduit through which the synergy of causation and effectuation translates into tangible innovation outcomes, by enabling founding teams to perceive where strategic action is needed and commit resources accordingly (Dollinger, 1984; Racat et al., 2024).

By engaging in boundary-spanning behaviors, founding teams become adept resource orchestrators, effectively connecting external inputs with internal innovation processes. This capability mitigates the inherent resource limitations of new ventures by providing access to complementary knowledge and assets (Leicht-Deobald et al., 2025), thereby reducing innovation risks, and optimizing resource allocation efficiency (Nason et al., 2019; Read et al., 2009). Moreover, boundary-spanning may help accelerate the identification and exploitation of emergent opportunities, ensuring that the adaptive flexibility of effectuation and the strategic foresight of causation are fully

harnessed (Reymen et al., 2015; Yan et al., 2020). In this way, boundary-spanning serves as a pivotal mechanism that mediates the relationship between integrated entrepreneurial decision-making logics and innovation performance.

Based on this theoretical synthesis, we propose the following hypothesis:

Hypothesis 2: The boundary-spanning behavior of the founding team mediates the relationship between the synergy of causation and effectuation and new venture innovation.

The moderating effect of founding team trust

In the entrepreneurial context, founding teams face the complex challenge: reconciling diverse decision-making preferences, aligning strategic priorities, and collectively orchestrating resources to drive innovation (Galkina et al., 2022; Karhu & Ritala, 2020). Trust, defined as shared generalized perceptions of trust that team members have in their fellow teammates, plays a critical role in this process. It fosters open communication, collaborative engagement, and effective coordination (De Jong & Elfring, 2010). Trust may create favorable conditions for the synergy between causation and effectuation to unfold, suggesting that it can amplify this synergy and facilitate boundary-spanning behaviors that are essential for innovation.

High levels of trust within founding teams cultivate an environment where members feel comfortable sharing diverse perspectives and tacit knowledge (Blatt, 2009). This openness is particularly vital when navigating the inherent tensions between structured planning in causation and adaptive experimentation in effectuation. Trust emboldens team members to embrace the uncertainty and risk-taking central to effectuation, knowing that their contributions will be valued (Chen et al., 2021; Dirks & Ferrin, 2001), while simultaneously fostering the mutual accountability needed to align efforts through causation. For example, a trusted team might confidently experiment with a novel partnership (effectuation) while ensuring it aligns with a strategic resource plan (causation), thereby harmonizing causation and effectuation with minimal friction.

This trust-based collaborative dynamic offers several benefits. First, it reduces interpersonal barriers like mistrust or miscommunication, which could otherwise hinder the synergy of causation and effectuation (Costa et al., 2018; Simons & Peterson, 2000). Second, it accelerates decision-making by fostering a shared understanding of goals and roles, enabling the team to swiftly adapt to emerging opportunities or challenges (Joshi et al., 2009). Third, trust enhances the coordination of boundary-spanning activities, as team members are more willing to collectively engage with external stakeholders, secure resources, and exchange knowledge (Collewaert & Sapienza,

2016). In essence, trust acts as a lubricant, facilitating the smooth interplay between causation's deliberate foresight and effectuation's opportunistic flexibility.

Therefore, we propose that the trust among the founding team amplifies the relationship between the synergy of causation and effectuation and boundary-spanning behavior. When trust is high, teams can more effectively leverage their integrated entrepreneurial decision-making logics to scan external environments, build strategic partnerships, and reconfigure resources—key facets of boundary-spanning that drive innovation. Conversely, low trust may stifle this synergy, as reluctance to share ideas or collaborate could weaken the team's ability to bridge internal processes with external opportunities. In essence, trust acts as a moderating force, intensifying the link between causation–effectuation synergy and boundary-spanning.

Based on this theoretical reasoning, we propose the following hypothesis:

Hypothesis 3: Founding team trust positively moderates the relationship between the synergy of causation and effectuation and team boundary-spanning behavior.

Moderated mediation model of founding team trust

Building on the reasoning above, boundary-spanning behaviors may play a crucial role in translating the synergy between causation and effectuation into tangible innovation outcomes. However, the effectiveness of this effect may depend on the level of trust within founding teams. Existing studies suggest that trust, characterized by mutual confidence and reliability, can foster a collaborative environment within teams, enabling the integration of entrepreneurial decision-making logics and the effective execution of boundary-spanning activities (De Jong et al., 2016; Grossman & Feitosa, 2018).

High levels of team trust can act as a critical boundary condition that influences the strength of boundary-spanning behaviors, thus, in turn, influencing the relationship between causation–effectuation synergy and innovation. First, trust strengthens the relationship between the synergy of causation and effectuation and boundary-spanning by promoting open communication and coordinated action (Liu et al., 2025). When trust is present, team members are more likely to align structured resource strategies with opportunistic experimentation, enabling engagement with external stakeholders to secure knowledge and resources (Büchel et al., 2013). Second, trust enhances the effectiveness of boundary-spanning itself, as it encourages rapid decision-making and adaptive collaboration with external partners, potentially

improving the team’s ability to translate inputs into innovative outputs (Olson et al., 2007).

In contrast, low trust may disrupt this mediated pathway, as interpersonal tensions or reluctance to collaborate could weaken the synergy of causation and effectuation, hindering boundary-spanning efforts. Trust acts as a moderating force that not only bolsters the direct link between entrepreneurial decision-making logics and boundary-spanning but also amplifies the indirect effect of this synergy on innovation through boundary-spanning behaviors. For instance, a high-trust team might swiftly pivot to exploit an unexpected partnership (effectuation) while ensuring it aligns with a strategic plan (causation), leveraging boundary-spanning to integrate external insights. This results in greater innovation compared to a low-trust team. Therefore, we propose that founding team trust shapes the strength of the mediated relationship between decision-making synergy and innovation.

Based on this theoretical synthesis, we propose the following hypothesis:

Hypothesis 4: Founding team trust moderates the indirect positive effect of the synergy of causation and effectuation on new venture innovation via boundary-spanning behavior.

Figure 1 presents our research framework.

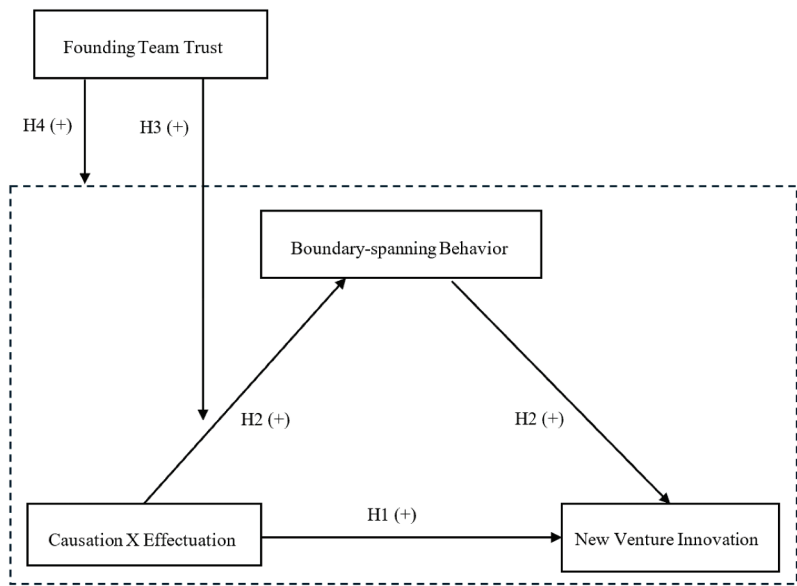


Figure 1. Research framework.

Methodology

Sample and data collection

We collected data from founding teams of new ventures, which were independently held and eight years old or less (McDougall et al., 1994; Miller & Camp, 1985), in China from 2016 to 2017. The study context reflects China's dynamic entrepreneurial ecosystem, characterized by strong government support through initiatives like "Mass Entrepreneurship and Innovation" (State Council, 2015), a thriving venture capital market that ranked second globally in startup funding, and dense industrial clusters benefiting from specialized infrastructure and knowledge spillovers (Zhang & Li, 2010).

The data collection followed a multi-wave survey (three waves with an interval of three months), with temporal separation between independent and dependent variables to reduce common method bias and ensure data quality (Podsakoff et al., 2003; Shirokova et al., 2020; Vardaman et al., 2018). We employed a rigorous, multi-stage approach to questionnaire design to ensure reliability and validity. First, we used established scales and implemented a translation-back-translation process (Brislin, 1970), consulting local entrepreneurship scholars to ensure contextual appropriateness. Second, we conducted a pilot study with 10 founding teams, refining item wording based on feedback and eliminating redundant items to mitigate respondent fatigue (Brownell et al., 2024). The pilot study samples were excluded from the final analysis to maintain data integrity. Additionally, considering that an overly long questionnaire may lead to respondent fatigue, reducing data quality and response rates, we followed Smolka et al. (2018) to remove the items within the same dimension that conveyed similar meanings and had low factor loadings.

We initially identified 400 new ventures from business directories, technology parks, and incubators distributed mostly in five regions of China: Beijing, Shanghai, Guangdong, Hubei, and Sichuan. These provinces are among the most economically advanced areas in China, characterized by high levels of entrepreneurship and innovation and intense market competition. The first-wave survey was delivered by contacting founding CEOs through personalized emails, explaining the research objectives, guaranteeing confidentiality, and promising customized feedback reports upon completion of the study. The questionnaire collected firm-level information (for example, establishment year, employee count, industry classification) and a complete roster of founding team members, which served as the foundation for subsequent waves, ultimately yielding 118 participating ventures. Three months later, we sent the second-wave questionnaire to all founding team members (except the CEO), collecting the individual-level data (for example, entrepreneurial experience, educational background, demographics) and team-level constructs, including causation and effectuation (independent variables), boundary-spanning behaviors (mediator), and founding team trust

(moderator). Finally, we conducted the third wave to ask the CEO to assess firm-level outcomes, including venture innovation (dependent variable).

After excluding 18 ventures that withdrew in the next two waves and nine with incomplete responses, we ultimately retained 91 sample ventures with 495 questionnaires from 404 founding team members (91 CEOs and 313 members). Sample descriptive statistics are listed in Table 1. Nearly half (44, 48.35 percent) of the new ventures had fewer than 20 employees, and more than half (48, 52.75 percent) had been in existence for less than two years, representing the typical characteristics of “small and new” ventures. They are mainly concentrated in electronic information, biological medicine, new energy and materials, mechanical manufacturing, and other industries. The founding team size varied from 3 to 11 members, with an average of 4.40 members. Of the 404 founding team members, 27.48 percent were female, and 43.07 percent were under the age of 30, which is in line with the trend of “youthfulness” among entrepreneurs. Most of the team members had a bachelor’s degree or above (343, 84.90 percent) and prior work experience (351, 86.88 percent), while 12.87 percent of them had previous entrepreneurial experience.

Table 1. Sample profile.

Characteristics	Distribution	Frequency	Percentage
Firm level (new ventures)			
FirmSize (N = 91)	1–20	44	48.35
	21–50	27	29.67
	51–100	11	12.09
	101 or above	9	9.89
FirmAge (N = 91)	0–2	48	52.75
	2–5	33	36.26
	5–8	10	10.99
Industry (N = 91)	New Medicine and Biotechnology Industry	8	8.79
	New Energy and New Materials Industry	7	7.69
	Mechanical Manufacturing Industry	4	4.40
	Electronic Information Industry	43	47.25
	Other Industries (Precision Instrument Industry)	29	31.87
Team level (founding teams)			
TeamSize (N = 91)	3	6	6.59
	4	60	65.93
	5	13	14.29
	6	8	8.79
	7 or above	4	4.40
Individual level (founding team members)			
Gender (N = 404)	Male	293	72.52
	Female	111	27.48
Age (N = 404)	30 or younger	174	43.07
	31–35	97	24.01
	36–40	69	17.08
	41 or older	64	15.84
Education (N = 404)	Less than Bachelor	61	15.10
	Bachelor	192	47.52
	Master	105	25.99
	PhD	46	11.39
Entrepreneurial experience (N = 404)	1	52	12.87
	0	352	87.13
Working experience (N = 404)	1	351	86.88
	0	53	13.12

Measurements

We used previously validated multi-item scales to measure each of the constructs contained in this study. All items were measured on a 7-point Likert scale (“1” = completely disagree, “7” = completely agree), with details for each scale provided in Appendix [Table A1](#).

New venture innovation

Drawing on Calantone et al. (2002), we defined new venture innovation as the willingness of a new venture to adopt new ideas, processes, or products, as well as to embrace change and implement new knowledge. We used a six-item scale to measure the level of innovation in new ventures. Two items were removed due to standardized estimates below 0.5 and also having the lowest factor loadings in the reference scale (Calantone et al., 2002; Hair et al., 2010), resulting in a final four-item scale.

The synergy of causation and effectuation

Based on the context of this study and drawing on the methods of Smolka et al. (2018) and Yu et al. (2018), we used the cross-term of causation and effectuation to measure the synergy. Causation comprised seven items mainly involving deterministic goals, maximizing expected returns, business planning, and competitive analyses. Effectuation comprised 13 items with four dimensions: experimentation, affordable loss, flexibility, and pre-commitment (Chandler et al., 2011). Following Smolka et al. (2018), we removed one item that was essentially similar in meaning, to make the questionnaire relatively concise and to ensure that the respondents could focus sufficiently. Before assessing the validities, we conducted a factor loading analysis separately for the causation and effectuation scales, which originally consisted of 7 and 12 items, respectively. Based on this analysis, we removed one item from the causation scale and two items from the effectuation scale due to underperformance (factor loading < 0.5). Additionally, we found that these removed items exhibited partial semantic overlap with other items in the scale. Consequently, the final analysis included 6 items for causation and 10 items for effectuation.

Boundary-spanning behavior of the founding team

Boundary-spanning behavior of the founding team, as measured by Ancona and Caldwell (1992), was a mediating variable comprised of three dimensions: ambassador behavior (twelve items), coordination behavior (five items), and scouting behavior (four items).

Founding team trust

Founding team trust was measured using a five-item scale developed by De Jong and Elfring (2010).

Control variables

We controlled for a set of variables that might contribute to the explanation of new venture innovation with reference to previous literature (Deligianni et al., 2022), including entrepreneurial firm attributes such as *FirmSize* (the number of employees, ln), *FirmAge* (the years that a new venture had been established), *Industry* (biological medicine industry, new energy and materials industry, machinery manufacturing industry, electronic information industry, and others). We also considered founding team-level attributes (Alzamora-Ruiz et al., 2021; Ruiz-Jiménez et al., 2021), including *TeamSize* (the number of founding team members), *Age* (the average age of founding team members), *Education* (the proportion of founding team members with a bachelor's degree or above; "1" = senior high school or below, "2" = junior college, "3" = Bachelor's, "4" = Master's, "5" = PhD), *Entrepreneurial experience* and *Working experience* (the proportion of founding team members with previous entrepreneurial/working experience; "1" = YES, "0" = NO), and *CEO gender* ("1" = Male; "0" = Female).

Reliability and validity

We assessed the reliability and validity of our constructs through multiple tests, with detailed results presented in Table 2. Cronbach's alpha values ranged from 0.82 to 0.94, and composite reliability (CR) ranged from 0.82 to 0.94, surpassing the recommended threshold of 0.7 (Cronbach, 1951; Hair et al., 2010). All factor loadings exceeded the recommended minimum of 0.5 (Hair et al., 2010), supporting convergent validity. Additionally, average variance extracted (AVE) values were above the acceptable threshold of 0.5 for four constructs (Hulland, 1999), while boundary-spanning behavior achieved an acceptable AVE of 0.44, collectively confirming convergent validity. Finally, discriminant validity was confirmed following Fornell and Larcker's (1981) criterion, the square roots of AVE (shown in Table 3 diagonals) exceeded

Table 2. Reliability, validity and aggregate test.

	Cronbach's alpha	CR	AVE	ICC (1)	ICC (2)	Rwg
New venture Innovation	0.82	0.82	0.54	–	–	–
Causation	0.88	0.89	0.57	0.27	0.56	0.94
Effectuation	0.92	0.93	0.56	0.36	0.66	0.98
Team boundary-spanning behavior	0.94	0.94	0.44	0.45	0.74	0.95
Founding team trust	0.88	0.88	0.61	0.53	0.80	0.79

Table 3. Correlations, means, and standard deviations.

Variables	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13
1. FirmSize (ln)	3.24	0.96	—												
2. FirmAge	2.42	1.57	0.45***	—											
3. TeamSize	4.44	1.14	0.45***	0.23**	—										
4. Age	33.06	5.39	0.53***	0.46***	0.25**	—									
5. Education	0.84	0.28	0.45***	0.07	0.18*	0.18*	—								
6. Entrepreneurial experience	0.12	0.16	0.13	0.13	0.22**	0.28***	0.15	—							
7. Working experience	0.86	0.23	0.16	0.00	0.15	0.12	−0.13	−0.13	—						
8. CEO gender	0.88	0.33	0.12	0.18*	0.05	0.13	0.15	−0.05	0.18*	—					
9. New venture innovation	4.84	0.98	−0.48***	−0.31***	−0.26**	−0.35***	−0.51***	−0.12	0.26**	−0.22**	—				
10. Causation	5.29	0.58	−0.33***	−0.29***	0.09	−0.35***	−0.27**	0.02	0.10	−0.15	0.53***	—			
11. Effectuation	5.39	0.50	−0.20*	−0.39***	0.19*	−0.33***	−0.09	−0.04	−0.08	−0.12	0.38***	0.75	—		
12. Boundary-spanning behavior	4.81	0.54	−0.01	−0.14	−0.15	−0.11	0.06	−0.22**	0.05	0.13	0.33***	0.25**	0.17	0.66	—
13. Founding team trust	5.31	0.68	−0.50***	−0.33***	−0.18*	−0.48**	−0.28***	−0.14	0.02	−0.02	0.44***	0.55***	0.46***	0.18*	0.78

N = 91; SD = standard deviation; The diagonal value is the square root of AVE.

+ $p < .10$; * $p < .05$; ** $p < .01$, *** $p < .001$.

correlations with other constructs. We also conducted confirmatory factor analysis (CFA), which indicated good model fit ($\chi^2/df = 1.13$; RMSEA = 0.04; CFI = 0.94; TLI = 0.94). Overall, our measures demonstrate strong reliability and validity.

Common method variance and aggregation to the team level

Common method biases assessment

We adopted both procedural and statistical measures to mitigate common method variance (Podsakoff et al., 2003). Procedurally, we employed a multi-wave data collection approach with a temporal separation of approximately three months between the measurement of independent and dependent variables, reducing potential biases inherent in cross-sectional designs (Shirokova et al., 2020; Taghvaei & Talebi, 2023). Additionally, questionnaires were assigned unique identification numbers to match respondents confidentially, and reverse-coded items were included to minimize response pattern bias. Statistically, we conducted Harman's single-factor test. The results revealed seven distinct factors, with the first factor explaining only 28.02 percent of the total variance—significantly below the recommended threshold of 50 percent. Thus, common method variance is unlikely to pose a substantial threat to our findings.

Aggregation adequacy testing

Before aggregating individual-level data into founding team-level variables, we computed well-accepted statistical indexes such as the Index of Within-Group Interrater Agreement (*Rwg*) (James et al., 1993) and Interclass Correlation Coefficient (*ICC (1)* and *ICC (2)*). Typically, when *Rwg* and *ICC (1)* are higher than the thresholds of 0.7 and 0.12, respectively, and *ICC (2)* is higher than the *ICC (1)* value (Bliese, 2000), the data are considered adequately reliable to be averaged to the team level and reflect team characteristics of respondents. All individual-level data were in accordance with the required criteria for aggregation at the founding team level. The details are listed in Table 2.

Results

Table 3 reports the number of observations, mean, and standard deviation of the sample. Additionally, the variance inflation factors (VIFs) of all models were significantly lower than the common criterion of 10, with values ranging from 1.58 to 3.70, thus eliminating the risk of multicollinearity (Neter et al., 1990).

The variables involving the interaction product were first mean-centered and then multiplied. We followed the hierarchical regression method to investigate the main and mediating effects. In Table 4, Model 1 was the initial

Table 4. Results of regression analysis testing for new venture innovation and results of the mediation analysis.

	New Venture Innovation			Boundary-Spanning Behavior		New Venture Innovation	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
Constant	6.52*** (0.59)	1.21 (1.04)	0.96 (0.98)	5.41*** (0.51)	3.18*** (0.81)	2.69*** (0.72)	-0.47 (0.95)
FirmSize (ln)	-0.30* (0.12)	-0.27* (0.11)	-0.24* (0.10)	0.08 (0.09)	0.14+ (0.08)	-0.36*** (0.10)	-0.30** (0.09)
FirmAge	-0.08+ (0.05)	0.01 (0.05)	-0.04 (0.05)	-0.07+ (0.04)	-0.09* (0.04)	-0.03 (0.05)	-0.00 (0.05)
TeamSize	-0.04 (0.07)	-0.14+ (0.09)	-0.09 (0.07)	-0.04 (0.07)	-0.04 (0.07)	-0.01 (0.06)	-0.07 (0.06)
Age	-0.01 (0.02)	0.02 (0.02)	0.02 (0.01)	-0.02 (0.01)	-0.00 (0.01)	0.01 (0.01)	0.02 (0.01)
Education	-0.98** (0.30)	-0.67** (0.21)	-0.81*** (0.21)	-0.11 (0.20)	-0.06 (0.20)	-0.90** (0.27)	-0.78*** (0.19)
Entrepreneurial experience	0.45 (0.40)	0.42 (0.39)	0.30 (0.35)	-0.48 (0.42)	-0.67+ (0.37)	0.78* (0.33)	0.60 (0.37)
Working experience	1.45*** (0.28)	1.48*** (0.21)	1.33*** (0.22)	-0.18 (0.23)	-0.38+ (0.21)	1.58*** (0.26)	1.50*** (0.22)
CEO gender	-0.53** (0.17)	-0.44*** (0.12)	-0.36** (0.13)	0.19 (0.16)	0.31** (0.11)	-0.67*** (0.12)	-0.50*** (0.14)
New Medicine and Biotechnology Industry	0.02 (0.35)	-0.50+ (0.28)	-0.59** (0.21)	0.46 (0.28)	0.19 (0.19)	-0.31 (0.24)	-0.68** (0.20)
New Energy and Materials Industry	-1.06** (0.39)	-1.08*** (0.28)	-1.10*** (0.25)	0.34 (0.22)	0.27 (0.17)	-1.30*** (0.28)	-1.22*** (0.20)
Mechanical Manufacturing Industry	0.64 (0.49)	0.52 (0.36)	-0.00 (0.23)	1.20*** (0.27)	0.73*** (0.14)	-0.21 (0.34)	-0.33 (0.21)
Electronic Information Industry	-0.12 (0.20)	-0.22 (0.15)	-0.21 (0.15)	0.21 (0.14)	0.18 (0.12)	-0.27 (0.18)	-0.29+ (0.15)
Causation		0.38* (0.15)	0.29* (0.14)		0.27+ (0.14)		0.17 (0.15)
Effectuation		0.45** (0.15)	0.51*** (0.14)		0.02 (0.14)		0.50*** (0.14)
Causation X Effectuation			0.83*** (0.17)		0.61*** (0.14)		0.55*** (0.16)
Boundary-spanning behavior						0.71*** (0.10)	0.45*** (0.12)
R ²	0.57	0.67	0.72	0.28	0.44	0.68	0.76
Adjusted R ²	0.50	0.61	0.67	0.17	0.33	0.62	0.70
F	13.31	28.31	29.84	2.93	15.73	26.74	38.01

N = 91; Unstandardized coefficients shown, with robust standard errors (The same applies in the following tables).
 †*p* < .10; **p* < .05; ***p* < .01, ****p* < .001.

model with only control variables, and Models 2–3 successively introduced causation and effectuation, and the interaction term. Model 2 shows that causation and effectuation are positively related to the innovation of new ventures ($\beta = 0.38$, $p < .05$; $\beta = 0.45$, $p < 0.01$; Model 2). Then, we included causation, effectuation, and their interaction term in Model 3, and found support for H1. The synergistic effect of causation and effectuation on the innovation of new venture is positive and significant ($\beta = 0.83$, $p < .001$; Model 3 and Figure 2). After standardizing the coefficients, we also confirmed causation and effectuation can be seen as mutually reinforcing (synergy): the sum of the direct effects and the interaction term (0.70; Model 3) is greater

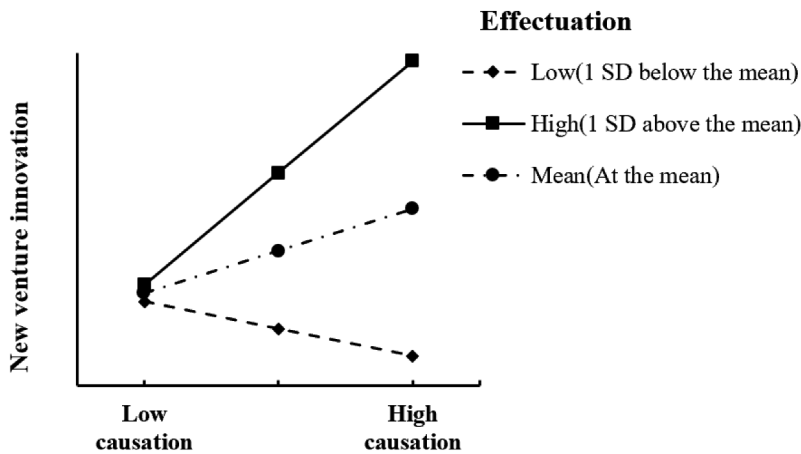


Figure 2. Interaction graph of causation and effectuation.

than the sum of the direct effects (0.46; Model 2). Figure 2 illustrates the interaction effects of different levels of causation and effectuation on innovation. The results indicate that innovation is maximized under a high-causation, high-effectuation configuration, suggesting a positive effect when both causation and effectuation are simultaneously leveraged. H2 predicted that boundary-spanning behavior mediates the relationship between synergy and innovation. The synergistic use of causation and effectuation is positively related to the founding team’s boundary-spanning behavior ($\beta = 0.61$, $p < .001$; Model 5), and this behavior is positively associated with the innovation of new ventures ($\beta = 0.71$, $p < .001$; Model 6). When we simultaneously included the interaction term and the mediator variable in Model 7, their coefficients were both significantly positive ($\beta = 0.55$, $p < .001$; $\beta = 0.45$, $p < .001$; Model 7). This confirms H2. We also used Bootstrap to re-test the mediating role and reinforce the conclusion of H2. The indirect effect of the synergistic use of causation and effectuation on innovation through the founding team’s boundary-spanning behavior is significantly positive, as the confidence interval does not include zero ($\beta = 0.31$, 95 percent CI [0.07, 0.58]). Moreover, the total and direct effects are 0.83 and 0.53, respectively.

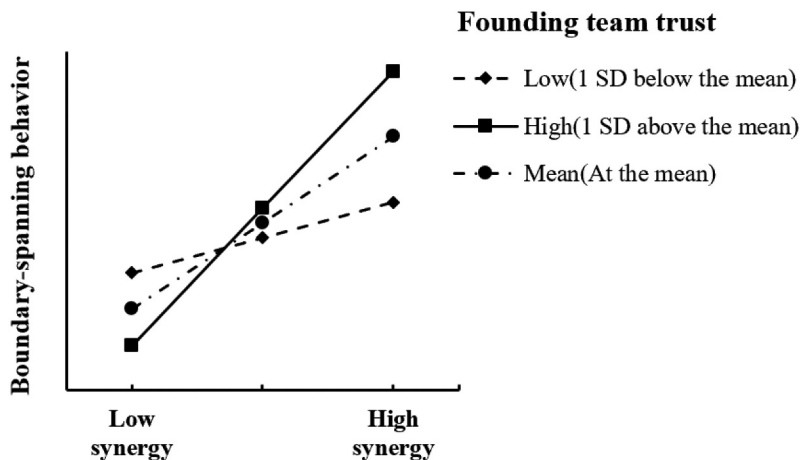
To test H3 and H4, we used a first-stage moderated mediation by introducing founding team trust as a moderator to the relationship between the synergy of causation and effectuation and team boundary-spanning behavior. Table 5 presents the results of the moderating effects, showing that the coefficient of the interaction term is significant ($\beta = 0.55$, $p < .05$; Model 1). Looking closer at the conditional effects, the relationship between the synergy and boundary-spanning behavior is positive but non-significant when in a low trust atmosphere (1 SD below the mean; $\beta = 0.26$, 95 percent CI [−0.21;0.72]). In contrast, this effect is positive and significant when in an average trust atmosphere (at the mean; $\beta = 0.63$, $p < .001$; 95 percent CI [0.28;0.98]), and is

Table 5. Results of the moderated mediation analysis on new venture innovation.

	Boundary-Spanning Behavior				New Venture Innovation			
	Model 1				Model 2			
	<i>B</i>	<i>SE</i>	95% <i>CI</i>		<i>B</i>	<i>SE</i>	95% <i>CI</i>	
Constant	4.29*	1.01	2.53	6.44	−0.35	1.14	−2.61	1.92
FirmSize (ln)	0.16 ⁺	0.09	−0.03	0.31	−0.30**	0.11	−0.49	−0.06
FirmAge	−0.07	0.04	−0.15	0.02	0.01	0.06	−0.10	0.14
TeamSize	−0.04	0.09	−0.28	0.06	−0.08	0.09	−0.32	0.04
Age	0.00	0.01	−0.03	0.02	0.02	0.02	−0.02	0.05
Education	−0.08	0.22	−0.49	0.39	−0.81***	0.23	−1.31	−0.41
Entrepreneurial experience	−0.76 ⁺	0.39	−1.56	0.01	0.72	0.45	−0.22	1.57
Working experience	−0.39	0.25	−0.91	0.07	1.53***	0.27	0.97	2.05
CEO gender	0.29 ⁺	0.15	−0.03	0.56	−0.53**	0.16	−0.85	−0.23
New Medicine and Biotechnology Industry	0.12	0.22	−0.22	0.66	−0.67	0.29	−1.30	−0.14
New Energy and Materials Industry	0.24	0.21	−0.11	0.73	−1.21***	0.25	−1.60	−0.61
Mechanical Manufacturing Industry	0.77***	0.22	0.46	1.22	−0.36	0.28	−0.87	0.26
Electronic Information Industry	0.18	0.13	−0.09	0.43	−0.32 ⁺	0.17	−0.64	0.02
Causation	0.17	0.14	−0.12	0.45	0.12	0.19	−0.23	0.51
Effectuation	−0.07	0.15	−0.34	0.24	0.53**	0.17	0.20	0.88
Causation X Effectuation	0.63***	0.17	0.22	0.88	0.53*	0.20	0.11	0.92
Founding team trust	0.05	0.11	−0.18	0.25				
Causation X Effectuation X Founding team trust	0.55*	0.23	0.12	1.01				
Boundary-spanning behavior					0.50**	0.15	0.17	0.78
Conditional effect/Conditional indirect effect								
Low team trust (1 SD below the mean)	0.26	0.23	−0.21	0.72	0.13	0.14	−0.16	0.35
Medium team trust (at the mean)	0.63	0.18	0.28	0.98	0.32	0.13	0.05	0.56
High team trust (1 SD above the mean)	1.00	0.24	0.52	1.48	0.50	0.19	0.12	0.88

N = 91; Variables for interaction terms are grand mean centered; all estimates for the moderated mediation model are tested for significance using bias-corrected 95% confidence intervals from 5,000 bootstrap samples; Index of moderated mediation = 0.27 and 95% CI [0.03; 0.61].

†*p* < .10; **p* < .05; ***p* < .01; ****p* < .001.

**Figure 3.** Boundary-spanning behavior as a function of the synergy and founding team trust.

more positive and significant when in a high trust atmosphere (1 SD above the mean; $\beta = 1.00$, $p < .001$; 95 percent CI [0.52;1.48]; see the bottom of Table 5), thereby corresponding to H3. We further computed and visualized the interaction effects in Figure 3. The plots clearly show that for founding teams with

high trust, the corresponding lines are steeper than others, indicating that trust has a strong positive impact on the synergy. In Model 2, the index of moderated mediation is positive, and the 95 percent confidence intervals do not include zero (Index = 0.27; 95 percent CI [0.03; 0.61]), indicating that the indirect synergistic effect on innovation through team boundary-spanning behavior is positively moderated by founding team trust. More specifically, the conditional indirect effects show that the indirect effect of the synergistic use of causation and effectuation on innovation through boundary-spanning behavior increases as team trust grows (Low 1 SD below the mean; $\beta = 0.13$, 95 percent CI [-0.16; 0.35]; at the mean; $\beta = 0.32$, 95 percent CI [0.05; 0.56]; High 1 SD above the mean; $\beta = 0.50$, 95 percent CI [0.12; 0.88]; see the bottom of Table 5). Therefore, these results support H4.

Further analysis

To demonstrate the robustness of our findings, we used the following robustness checks, with the data detailed in Appendix Table A2. First, we removed all control variables. The regression result between the independent and dependent variables shows a stable and significant positive relationship ($\beta = 0.99$, $p < .05$; Model 1). Then, we used an alternative measure of innovation to replace the dependent variable. Specifically, we used the six-item survey scale of Guan and Liu (2016) to assess explorative innovation. The data were obtained from the founding teams during the second wave of our survey collection and have good reliability and validity. The results of robustness testing using the hierarchical regression method were consistent with the results reported above ($\beta = 0.46$, $p < .01$; Model 4), thus lending credence to the robustness of our findings.

Stam and Elfring (2008) stated that smaller firms often have resource constraints, which limit their ability to seize opportunities and, as a result, affect their capacity to be innovative. Given the limited access to innovation resources, larger new ventures are more likely to be capable of building external networks, thereby acquiring the necessary resources for innovation. This may introduce sample selection bias. To address potential sample selection bias, we applied Heckman's (1979) two-stage model to reexamine the relationship between synergy and new venture innovation. In the first stage, a Probit regression model was used, and the dependent variable (Treat) equaled one if the new venture's employee number exceeded the industry median. We used the industry-level mean of the independent variable as an instrumental variable (IV) and computed the inverse Mills ratio (IMR). The IV was positively associated with Treat ($\beta = 27.08$, $p < .05$; Model 5), supporting the validity of the chosen instrument. In the second step, IMR was brought into the model for regression, and the coefficient was not significant, indicating that there was no problem of sample selection. From the perspective of the

selection equation of new venture innovation, the impact of synergy on new venture innovation is consistent with the findings ($\beta = 0.74$, $p < .05$; Model 6), which further strengthens the robustness of the conclusion.

Discussion

New venture innovation is a complex process that requires founding teams to systematically integrate planned causation and flexible, adaptive effectuation. While prior research confirms that this synergy enhances venture performance, the mechanisms driving innovation outcomes and the conditions in which they work effectively remain underexplored. To address this gap, we employed resource orchestration theory to develop and test a model examining how the synergy of causation and effectuation fosters new venture innovation through founding team boundary-spanning behaviors, with founding team trust moderating this relationship. Based on data from 404 founding team members across 91 new ventures, our findings reveal compelling insights into these dynamics.

A central finding (Hypothesis 1) is that the synergy of causation and effectuation significantly boosts innovation by uniting causation's predictive, resource-efficient planning with effectuation's responsive, opportunity-driven adaptability (Alzamora-Ruiz et al., 2021). This dual approach enables founding teams to navigate uncertainty while optimizing resources, creating a balanced framework that enhances creative outcomes. Rather than operating in isolation, causation and effectuation reinforce each other, underscoring their complementary power as a catalyst for innovation. In addition, we tested the conditions under which synergy operates effectively, and the plotted slope graphs (Figure 2) show that innovation is maximized when both causation and effectuation operate at high levels simultaneously. Interestingly, when causation scores below the mean, effectuation doesn't compensate and may even have a negative effect, with higher effectuation weakening the boost to new venture innovation. This aligns with Smolka et al.'s (2018, p. 578) view: "The ability of effectuation processes to contribute to venture performance might therefore well be contingent upon the presence of at least a threshold level of causation processes, and the other way around."

Our mediation analysis (Hypothesis 2) further demonstrates that boundary-spanning behavior serves as a critical mechanism linking this synergy to innovation. By engaging in activities like seeking external resources, revising goals, and forging strategic partnerships, founding teams bridge internal capabilities with external opportunities (Gibson & Dibble, 2013; Marrone, 2010). This aligns with prior work suggesting that resource-scarce teams adopt boundary-spanning as a strategic response to uncertainty and internal limitations (Choi, 2002). However, such efforts are not without costs. For resource-constrained ventures, frequent boundary-spanning can

hinder innovation and incur risks, such as resource dilution, misalignment, and attention distraction (Faraj & Yan, 2009). Entrepreneurs often face conflicting goals and tight resource trade-offs (Long et al., 2023), requiring them continuously to prioritize attentional allocation (DeShon et al., 2004; Kanfer & Ackerman, 1989). Boundary-spanning is effortful and may directly compete with internally focused tasks for scarce attentional and organizational resources (Ancona, 1990; Choi, 2002). Marrone (2010) argued that although team boundary-spanning behaviors consistently target task goals, “these actions may or may not result in desirable consequences”, such as inefficient resource allocation and innovation setbacks (p. 914). A possible reason for the divergence between the risks outlined above and the positive effects observed in our study may lie in who engages in boundary-spanning activities and how these activities are distributed. In our context, it is the founding teams as a whole—serving as both strategic decision-makers and operational leaders—that engage in boundary-spanning behaviors, in contrast to prior research which typically focuses on individual employees as boundary-spanners (Keller & Holland, 2017; Marrone et al., 2007). This shared involvement may reduce the overburdening of any single member and enhance team coordination, thereby translating boundary-spanning efforts into performance gains rather than liabilities (Choi, 2002). This proactive external engagement magnifies the synergy’s impact, demonstrating that innovation depends not only on internal decision-making but also on the ability to orchestrate resources beyond organizational boundaries.

Equally important is the moderating role of founding team trust (Hypotheses 3 and 4). Prior research has demonstrated that trust plays a pivotal role in facilitating open communication, promoting information exchange, and resolving frictions commonly arising within a team (De Jong et al., 2016; Zheng, 2012). Trust is thus essential for managing tensions associated with integrating divergent entrepreneurial decision-making logics, as it increases team members’ willingness to accept stated disagreements (Simons & Peterson, 2000) and reach consensus, thereby ensuring efficient team operations (Peterson & Behfar, 2003). Based on this, our findings show that in a high-trust environment, founding teams that utilize both causation and effectuation are more willing to engage in boundary-spanning behaviors, which in turn fosters innovation. These results echo the broader recognition of trust as a cornerstone of effective team functioning (Costa et al., 2018), highlighting its strategic importance. In this sense, trust serves not only as a relational lubricant but also as a strategic enabler.

In sum, these findings illustrate how the causation–effectuation synergy drives innovation through boundary-spanning behavior, with trust acting as a linchpin that harnesses the positive synergy effect of causation and effectuation. Moreover, integrating entrepreneurial decision-making logics lays the

foundation for innovation. The effectiveness of this integration can be influenced by a team's ability to span boundaries and cultivate trust. This nuanced perspective offers a comprehensive understanding in dynamic contexts.

Theoretical contributions

Our study makes three significant contributions. First, we shift the focus of research on entrepreneurial decision-making from general firm performance outcomes (Smolka et al., 2018; Yu et al., 2018) to new venture innovation, a specific and theoretically important outcome. While prior research acknowledges that the synergy of causation and effectuation boosts venture-level results such as growth and profitability (Chen & Xu, 2022; Guo et al., 2016), the underlying mechanisms between such synergy and new venture innovation remain largely underexplored. By identifying boundary-spanning behavior as a mediating mechanism, we demonstrate how founding teams integrate structured planning from causation and adaptive flexibility from effectuation to orchestrate resources for innovation. As Dollinger (1984) noted, entrepreneurial boundary-spanning activities reflect both the perception of necessary environmental aspects and organizational resource commitment. Entrepreneurs translate signals from the external environment into strategic action precisely because, through boundary-spanning, they perceive where action is needed. Their investment of resources—such as time and expertise spent cultivating external relations or shaping firm image—reflects decision-making in action. In this regard, boundary-spanning serves as a cognitive-operational interface that translates the synergistic benefits of causation and effectuation into action (Racat et al., 2024). This reveals how synergy manifests through boundary-spanning behaviors, enriching our understanding of entrepreneurial decision-making dynamics (Chandler et al., 2011).

Second, we extend resource orchestration theory to entrepreneurial contexts characterized by resource scarcity, uncertainty, and external dependence that are prevalent (Baert et al., 2016; Nason et al., 2019). Unlike prior studies focusing on internal resource optimization in established firms, which emphasized hierarchical managerial coordination and strategic alignment (Carnes et al., 2022; Chadwick et al., 2015), we demonstrate that founding teams leverage boundary-spanning behaviors to access and integrate external resources. This dual reliance on structured and flexible decision-making redefines resource orchestration as a dynamic, externally oriented process in new ventures, expanding the scope and explanatory power within entrepreneurial contexts. Importantly, different dimensions of boundary-spanning behaviors correspond to distinct processes of resource orchestration. Specifically, ambassador behaviors help secure recognition and resource support from higher-level external actors, aligning with the structuring process. Coordination behaviors align workflows and integrate external feedback,

facilitating capability development and bundling. Scouting behaviors, through environmental scanning and information gathering, enable opportunity recognition and strategic choice, supporting both structuring and leveraging processes (Ancona & Caldwell, 1992). By operationalizing resource-focused actions through boundary-spanning behaviors (Sirmon et al., 2011), we enhance the theory's applicability in entrepreneurial settings, translating its abstract processes into concrete, team-level behavioral patterns.

Third, we enrich the team trust literature by reframing trust as a strategic moderator, transcending its traditional micro-level role in team dynamics (Dirks & Ferrin, 2001). Beyond facilitating routine cooperation, trust strategically moderates the integration of causation and effectuation, enhancing boundary-spanning efforts and innovation outcomes. By alleviating cognitive and coordination tensions inherent in blending causation and effectuation (Collewaert & Sapienza, 2016; Joshi et al., 2009), trust fosters a shared vision to guide how to structure, bundle, and leverage resources (Helfat, 2007). This novel perspective explains why some teams excel in resource orchestration while others falter, shedding light on the strategic significance of trust. Our findings also underscore that trust plays a particularly critical role in dynamic, resource-constrained entrepreneurial environments. While prior research—often grounded in stable organizational settings—has pointed out the potential risks of high team trust, such as “NIH syndrome” and bias on information checking (Chen & Wang, 2008; Katz & Allen, 1982), our study suggests that trust and boundary-spanning, when jointly activated, can effectively support innovation. By engaging in external scanning, coordination, and alliance-building, founding teams expose themselves to diverse perspectives and feedback loops that challenge internal biases and enhance adaptive capacity. By emphasizing the value of trust under entrepreneurial uncertainty, our study deepens the theoretical understanding of when and why trust fosters—rather than impedes—resource orchestration and innovation.

Practical implications

Our findings offer practical insights for founding teams striving to enhance innovation. First, founding teams should embrace the simultaneous use of causation and effectuation, blending structured analysis with agile responsiveness to orchestrate scarce resources effectively. Proactively engaging in boundary-spanning behaviors—such as forging alliances, monitoring competitors, and acquiring external knowledge—empowers teams to overcome resource constraints and market uncertainties, ultimately driving innovation. To operationalize this, founding teams should actively engage in boundary-spanning behaviors. In new ventures, this could involve designating specific team members to monitor competitor strategies, cultivating informal ties with upstream suppliers or downstream distributors, or participating in local

industry meetups and incubators to scout for partnership opportunities. Xiaomi, for example, leveraged co-development forums (for example, MIUI community) and online communities to access distributed knowledge and test ideas—showcasing how even resource-poor new ventures can integrate external inputs to refine and accelerate innovation.

Second, integrating these complementary yet conflicting distinct entrepreneurial decision-making logics requires managing inherent tensions. Cultivating high trust within founding teams is critical to this process. Trust fosters open communication, reduces conflict, and aligns divergent decision-making preferences. Practical steps like regular feedback sessions (for example, biweekly reflection meetings to align goals and address conflicts), transparent dialogue, and team-building activities can strengthen trust, enhance coordination, and sustain innovation over time.

Limitations and future prospects

Despite its valuable contributions, this study has several limitations that offer directions for future research. First, while our multi-wave data collection mitigates common method bias, the limited observation period restricts our ability to fully capture the dynamic evolution and sequential interplay between causation and effectuation. Innovation processes are inherently temporal, and the synergy between entrepreneurial decision-making logics may unfold differently over time. Future research employing extended longitudinal designs or in-depth qualitative case studies could provide richer insights into how causation and effectuation synergize over time, when tensions emerge, and how synergy is achieved or disrupted. In particular, more sophisticated analytical approaches—such as subgroup comparisons—could be used to explore different causation-effectuation configurations (for example, high-high, high-low), thereby identifying thresholds or asymmetries in their effects on innovation. In addition, given that our data were drawn exclusively from China—a context with distinct institutional frameworks and entrepreneurial ecosystem dynamics—the applicability of our findings to other settings remains an open question. Future studies should test the robustness of our results using larger, cross-national samples, which would help clarify the boundary conditions of our theoretical framework by considering cultural, regulatory, or market-specific factors.

Second, although our study is grounded in resource orchestration theory—which emphasizes structuring, bundling, and leveraging resources—the use of boundary-spanning behavior as a mediating mechanism may not fully capture this orchestration process, particularly regarding the strategic integration of external stakeholders. Future research could adopt more targeted measures to test this theoretical pathway more precisely. Additionally, we acknowledge that boundary-spanning in new ventures is

not without constraints. Allen et al. (1979) pointed out that effective boundary spanning involves two functions: interacting with external actors to acquire knowledge and engaging internally to integrate that knowledge within the organization. This implies that founding teams may face the challenge of balancing external exploration with internal coordination (Tushman & Scanlan, 2017). Given the diversity and ambiguity inherent in new venture's operations, boundary-spanning tasks are more complex and cognitively demanding. Limited information-processing capacity can hinder performance (Dollinger, 1984), and factors such as knowledge stickiness, inter-organizational distance, and (later) boundary-spanning phases further constrain effectiveness (Zobel et al., 2024). Future research could adopt a more critical perspective to examine the boundary conditions under which boundary-spanning behaviors facilitate—or potentially hinder—resource orchestration in entrepreneurial settings.

Third, although we acknowledge the potential for intra-team conflicts in the process of synergy, conflicts may arise at different levels—individual, intra-organizational, and inter-organizational—or even span multiple levels simultaneously (Galkina et al., 2022). A promising direction for future research would be to explore conflict resolution across these different levels to deepen our understanding of synergy. Additionally, trust encompasses multiple dimensions, including cognitive and affective forms, which may exert differential effects on entrepreneurial decision-making. Future studies should explore the nuanced roles of different trust forms and clarify the boundary conditions that shape their effectiveness. This will provide a more comprehensive understanding of trust's influence on managing tensions associated with entrepreneurial decision-making logic integration and innovation processes.

Conclusion

In response to growing interest in the synergy of entrepreneurial decision-making logics, this study offers a novel perspective on how founding teams navigate uncertainty and resource scarcity by strategically combining causation and effectuation. This synergy, achieved through the founding team's boundary-spanning behaviors, enhances innovation outcomes. Moreover, our study demonstrates that internal trust among founding team members fosters effective boundary-spanning behaviors, enabling them to coordinate external resources. By investigating the interplay between causation and effectuation, our study highlights both their synergistic and inherent tensions. This holistic approach advances the theory of entrepreneurial decision-making and resource orchestration, offering a deeper understanding of innovation in new venture settings.

Disclosure statement

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Appendix A

Table A1. Definition and sources of variables.

Item description	Standardized loadings
New Venture Innovation <i>Calantone et al., (2002)</i>	
Our company frequently tries out new ideas.	0.64
Our company seldom seeks out new ways to do things.	0.73
Our company is creative in its methods of operation.	0.74
Our company is often the first to market with new products and services.	0.81
Causation <i>Chandler et al., (2011)</i>	
We analyzed long-run opportunities and selected what we thought would provide the best returns.	0.68
We developed a strategy to best take advantage of resources and capabilities.	0.75
We organized and implemented control processes to make sure we met objectives.	0.76
We researched and selected target markets and did meaningful competitive analysis.	0.83
We had a clear and consistent vision for where we wanted to end up.	0.69
We designed and planned production and marketing efforts.	0.79
Effectuation <i>Chandler et al., (2011)</i>	
We experimented with different products and/or business models.	0.67
The product/service that we now provide is substantially different than we first imagined.	0.59
We tried a number of different approaches until we found a business model that worked.	0.84
We were careful not to commit more resources than we could afford to lose.	0.83
We were careful not to risk more money than we were willing to lose with our initial idea.	0.77
We were careful not to risk so much money that the company would be in real trouble financially if things didn't work out.	0.62
We were flexible and took advantage of opportunities as they arose.	0.78
We avoided courses of action that restricted our flexibility and adaptability.	0.82
We used a substantial number of agreements with customers, suppliers and other organizations and people to reduce the amount of uncertainty.	0.81
We used pre-commitments from customers and suppliers as often as possible.	0.69
Team Boundary-Spanning Behavior <i>Ancona & Caldwell, (1992)</i>	
Absorb outside pressures for the team so it can work free of interference.	0.59
Protect the team from outside interference.	0.56
Prevent outsiders from "overloading" the team with too much information or too many requests.	0.60
Persuade other individuals that the team's activities are important.	0.52
Scan the environment inside your organization for threats to the product team.	0.69
"Talk up" the team to outsiders.	0.66
Persuade others to support the team's decisions.	0.69
Acquire resources (for example, money, new members, equipment) for the team.	0.76
Report the progress of the team to a higher organizational level (for example, Investors and suppliers).	0.72
Find out whether others (for example, Investors and suppliers) in the company support or oppose your team's activities.	0.79
Find out information on your company's strategy or political situation that may affect the project.	0.78
Keep other groups in the company informed of your team's activities.	0.62
Resolve design problems with external groups.	0.69
Coordinate activities with external groups.	0.64
Procure things that the team needs from other groups or individuals in the company.	0.58
Negotiate with others for delivery deadlines.	0.76
Review product design with outsiders.	0.71
Find out what competing firms or groups are doing on similar projects.	0.55
Scan the environment inside or outside the organization for marketing ideas/expertise.	0.65
Collect technical information/ideas from individuals outside of the team.	0.66
Scan the environment inside or outside the organization for technical ideas/expertise.	0.67
Founding Team Trust <i>De Jong & Elfring, (2010)</i>	
I am able to count on my team members for help if I have difficulties with my job.	0.73
I am confident that my team members will take my interests into account when making work-related decisions.	0.65
I am confident that my team members will keep me informed about issues that concern my work.	0.88
I can rely on my team members to keep their word.	0.76
I trust my team members.	0.86

Table A2. Robustness testing.

	New Venture Innovation	Explorative Innovation			Treat	New Venture Innovation
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Constant	4.64*** (0.11)	4.88*** (0.51)	−0.08 (1.11)	−0.22 (1.07)	−19.15** (6.31)	1.63 (2.55)
FirmSize (ln)		0.14 (0.12)	0.18* (0.08)	0.20* (0.08)		0.14 (0.21)
FirmAge		−0.04 (0.05)	0.04 (0.04)	0.02 (0.04)	0.52* (0.24)	−0.01 (0.08)
TeamSize		0.03 (0.05)	−0.07+ (0.04)	−0.04 (0.04)	0.07 (0.21)	−0.06 (0.07)
Age		−0.00 (0.02)	0.02 (0.01)	0.02 (0.02)	0.31*** (0.08)	0.06 (0.03)
Education		0.18 (0.29)	0.47* (0.23)	0.39+ (0.23)	2.28* (0.99)	−1.11** (0.38)
Entrepreneurial experience		−0.74 (0.54)	−0.79+ (0.41)	−0.85* (0.40)	0.28 (1.09)	−0.09 (0.59)
Working experience		0.08 (0.31)	0.09 (0.26)	0.00 (0.25)	2.76* (1.40)	1.36+ (0.70)
CEO gender		−0.26 (0.19)	−0.17 (0.12)	−0.13 (0.13)	−0.12 (0.54)	−0.37 (0.26)
New Medicine and Biotechnology Industry		0.64* (0.31)	0.15 (0.22)	0.11 (0.19)	−7.24** (2.26)	−2.14*** (0.52)
New Energy and Materials Industry		−0.03 (0.26)	−0.05 (0.17)	−0.06 (0.16)	−2.76** (0.88)	−1.80*** (0.48)
Mechanical Manufacturing Industry		0.43* (0.21)	0.31+ (0.16)	0.02 (0.24)	−23.23* (9.10)	−1.66* (0.64)
Electronic Information Industry		0.25 (0.18)	0.16 (0.14)	0.17 (0.14)		−1.23** (0.33)
Causation			0.39* (0.17)	0.34* (0.17)	−0.52 (0.43)	−0.10 (0.23)
Effectuation			0.39* (0.19)	0.43* (0.17)	0.40 (0.58)	0.41 (0.25)
Causation X Effectuation	0.99* (0.38)			0.46** (0.15)	2.12 (1.29)	0.74* (0.29)
IV					27.08* (11.09)	
IMR						−0.26 (0.37)
N	91	91	91	91	91	43
R ²	0.11	0.23	0.48	0.52		0.83
Adjusted R ²	0.10	0.11	0.38	0.42		0.72
F	6.79	2.51	7.54	8.23		32.86

Unstandardized coefficients shown, with robust standard errors; † $p < .10$; * $p < .05$; ** $p < .01$, *** $p < .001$.